CSE4062 Introduction to Data Science and Analytics
Spring 2023

FINAL EXAM

Ethical Code

- You need to write your answers to a white A4 size papers by your handwriting ONLY!
- You need to write your full name, your number, department on the top of all of your answer papers.
- In the first page, after writing your name and number please write the following statement and sign:

"I am aware that any forms of cheating in this exam will result in a zero grade and a disciplinary investigation. I accept all rules and regulations regarding online exams. I give permission for the processing of my personal data as stated in the Clarification Text provided on the Faculty of Engineering website."

- **Each question is 25 points.**

SESSION 1

- Start **09:00**
- End **09:40**
- Upload to ues.marmara.edu.tr in 10 minutes by **09:50**

	A	B	C
A	50	10	20
B	5	150	5
C	10	10	200

1) Calculate the Precision, Recall, and F1 scores for each class (A, B, C) and their micro averages of precision and recall.

rows show actual class labels, P=TP/(TP+FP), R=TP/(TP+FN), F1=2FP/(F+P)

2) Assume Class A (Cancer), Class B (Normal), and Class C (Benign). Misclassifying a cancer patient as normal has a cost of 10, while misclassifying a normal patient as cancer has a cost of 2, and misclassifying a benign patient as cancer or normal has a cost of 1. Calculate the total cost of misclassifications for this model.

Q1 Answer:

Precision A = 50 / (50 + 5 + 10) = 0.769
B = 150 / (10 + 150 + 10) = 0.882
C = 200 / (20 + 5 + 200) = 0.888

Recall A = 50 / (50 + 10 + 20) = 0.625
B = 150 / (5 + 150 + 5) = 0.938
C = 200 / (10 + 10 + 200) = 0.909

F1 A = 2 * (50/65 * 50/80) / (50/65 + 50/80) = 0.689
B = 2 * (150/170 * 150/160) / (150/170 + 150/160) = 0.909
C = 2 * (200/225 * 200/220) / (200/225 + 200/220) = 0.898

Total TP = 50 + 150 + 200 = 400
Total FP = 10 + 5 + 10 + 10 + 20 + 5 = 60
Total FN = 10 + 20 + 5 + 5 + 10 + 10 = 60
Micro Avg P = 400 / (400 + 60) = 0.941 = Micro Avg R (because total FP = total FN)

Q2 Answer:

To calculate the total cost of misclassifications, we need to multiply the number of misclassifications between each pair of classes by their respective costs and sum the results.

Misclassifications:

Cancer (Class A) misclassified as Normal (Class B): 10 * 10 (cost) = 100

Normal (Class B) misclassified as Cancer (Class A): 5 * 2 (cost) = 10

Benign (Class C) misclassified as Cancer (Class A): 10 * 1 (cost) = 10

Benign (Class C) misclassified as Normal (Class B): 10 * 1 (cost) = 10

Total Cost of Misclassifications = 100 + 10 + 10 + 10 = 130

Therefore, the total cost of misclassifications for this model, considering the criticality of misclassifying a cancer patient as normal, is 130.

SESSION 2

- Start **10:00**
- End **10:35**
- Upload to ues.marmara.edu.tr in 10 minutes by **10:45**

A	B	C	D
1	1	1	1
1	1	0	1
1	0	0	1
1	0	1	1
0	1	1	0
0	0	1	0
0	0	0	0
0	1	0	?

Last row is test instance. Show all steps of your calculations!

3. D is the class attribute. Classify test instance using 3-nearest neighborhood algorithm. Use Manhattan distance.

d(x2,x1)=|x2[1] - x1[1]| + |x2[2] - x1[2]| + … + |x2[n] - x1[n]|
(Hint: do not include class attribute in your distance calculations)

4. D is the cluster id's of instances. Determine the cluster of test instance by calculating Manhattan distance to the cluster centroids.

ANSWER 3:
A: 1,1,1,1,0,0,0,0
B: 1,1,0,0,1,0,0,1
C: 1,0,0,1,1,1,0,0
D: 1,1,1,1,0,0,0,?



A	B	C	D
1	1	1	1
1	1	0	1
1	0	0	1
1	0	1	1
0	1	1	0
0	0	1	0
0	0	0	0
0	1	0	?

Manhattan Distance
d(1,test) = |1-0|+|1-1|+|1-0| = 2
d(2,test) = |1-0|+|1-1|+|0-0| = 1
d(3,test) = |1-0|+|0-1|+|0-0| = 2
d(4,test) = |1-0|+|0-1|+|1-0| = 3
d(5,test) = |0-0|+|1-1|+|1-0| = 1
d(6,test) = |0-0|+|0-1|+|1-0| = 2
d(7,test) = |0-0|+|0-1|+|0-0| = 1

Instances **2**, **5**, and **7** are closest to test instance.
Two of them **class 0**,
One of them **class 1**,
therefore I would classify the test instance as **0**

ANSWER 4:

Step 1: Calculate Cluster Centroids

We'll calculate the cluster centroids by finding the mean of each attribute for each cluster, excluding the last row and the cluster ID column (D).

Cluster One:

Mean of attribute A = (1 + 1 + 1 + 1) / 4 = 1 Mean of attribute B = (1 + 1 + 0 + 0) / 4 = 0.5 Mean of attribute C = (1 + 0 + 0 + 1) / 4 = 0.5

Cluster Two:

Mean of attribute A = (0 + 0 + 0) / 3 = 0 Mean of attribute B = (1 + 0 + 0) / 3 = 0.33 Mean of attribute C = (1 + 1 + 0) / 3 = 0.67

The cluster centroids are as follows:

Cluster One: (1, 0.5, 0.5)

Cluster Two: (0, 0.33, 0.67)

Step 2: Calculate Distance distance between the test instance (0, 1, 0) and each cluster centroid.

Cluster One: Manhattan distance = |0 - 1| + |1 - 0.5| + |0 - 0.5| = 1 + 0.5 + 0.5 = 2

Cluster Two:Manhattan distance = |0 - 0| + |1 - 0.33| + |0 - 0.67| = 0 + 0.67 + 0.67 = 1.34

Therefore, the test instance (0, 1, 0) is assigned to Cluster Two.